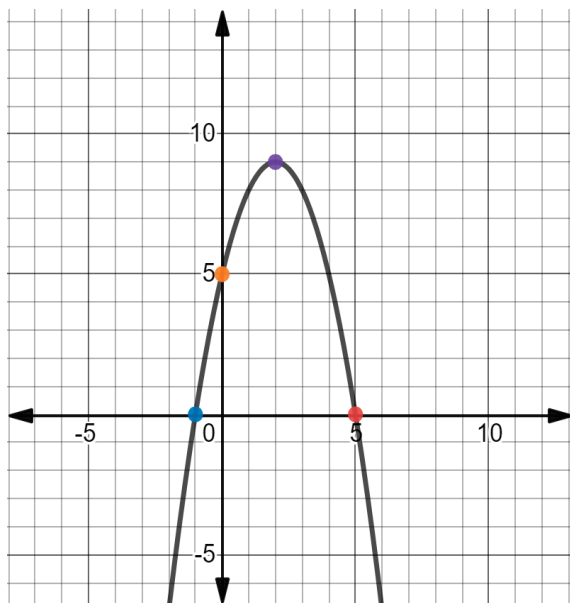




8.1.5 Wrap-Up: Cool Down

1. The graph of a quadratic equation is shown. Label the four colored points on the parabola with the ordered pair.
2. Use words, ordered pairs, or symbols to describe the location of each key feature on the given parabola.



Key Feature	Location of Key Feature
x -intercepts	
y -intercept	
vertex	
axis of symmetry	
increasing/decreasing	
positive/negative	



Unit 8, Lesson 2: More Key Features of Quadratic Functions



Benchmark

MA.912.AR.3.7 Given a table, equation, or written description of a quadratic function, graph that function, and determine and interpret its key features.



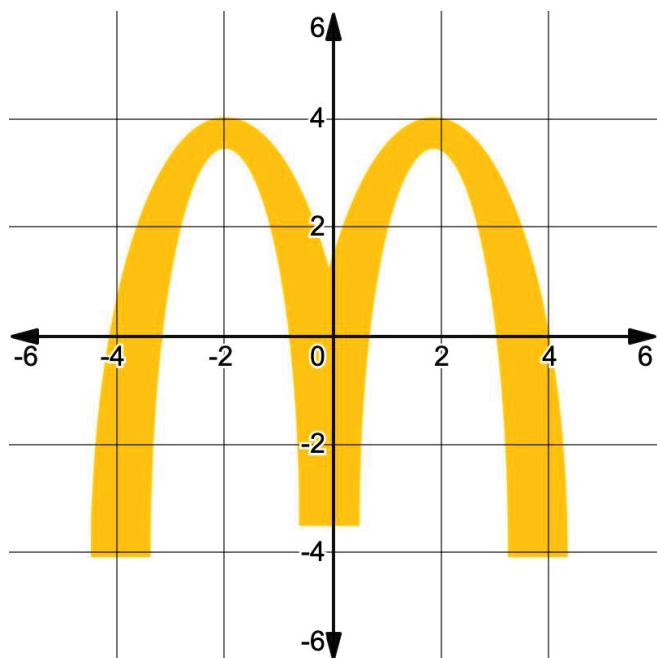
Learning Targets

- ☐ I can identify key features of a quadratic given a graph, a table, or an equation.
- ☐ I can graph a parabola given an equation or a table.

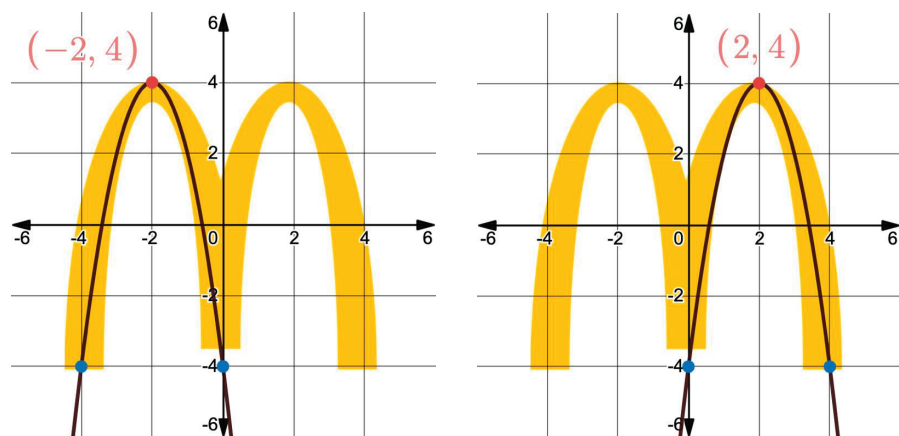


8.2.1 Warm-Up: Henry's Arches

1. Sketch a curve on the graph that fits one of the arches.



Henry used a computer program to draw curves that fit each of the arches. His curves are shown.



2. What do you notice about the two curves?

The computer program also gives Henry the equations and a table of value for the curves.

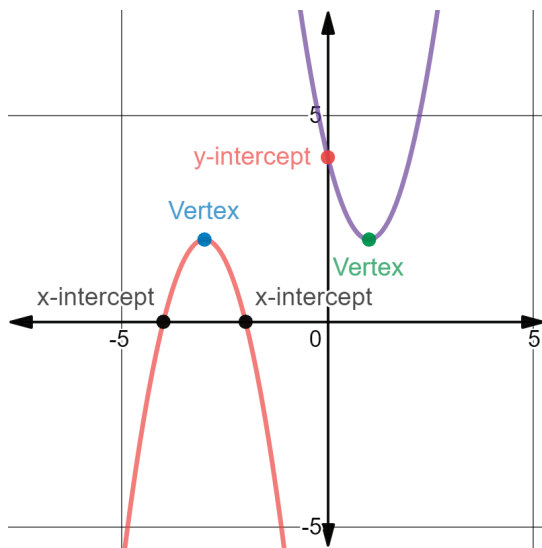
Left Arch		Right Arch	
$y = -2(x - (-2))^2 + 4$ OR $y = -2(x + 2)^2 + 4$		$y = -2(x - 2)^2 + 4$	
x	y	x	y
-3	2	-2	-28
-2	4	-1	-14
-1	2	0	-4
0	-4	1	2
1	-14	2	4
2	-28	3	2

3. What is one thing you notice about the tables of values?
4. Which key feature was the most useful in matching a quadratic to the arch?



8.2.2 Guided Instruction: More Key Features of Quadratics

Two different quadratic equations are shown on the coordinate grid.



The **end behavior** of a function refers to the appearance of the function's graph at the "ends," in other words, as the x -values get infinitely smaller or as the x -values get infinitely larger.

Use the definition of end behavior to complete each statement.

1. For the **red** parabola, as the x -values get infinitely **smaller**, the ends of the graph of the **red** parabola approach

☐ $-\infty$.
☐ ∞ .
2. For the **red** parabola, as the x -values get infinitely **larger**, the ends of the graph of the **red** parabola approach

☐ $-\infty$.
☐ ∞ .
3. For the **purple** parabola, as the x -values get infinitely **smaller**, the ends of the graph of the **purple** parabola approach

☐ $-\infty$.
☐ ∞ .
4. For the **purple** parabola, as the x -values get infinitely **larger**, the ends of the graph of the **purple** parabola approach

☐ $-\infty$.
☐ ∞ .



The **domain** is the complete set of possible values of the input of a function or relation. The domain may vary depending on the context.

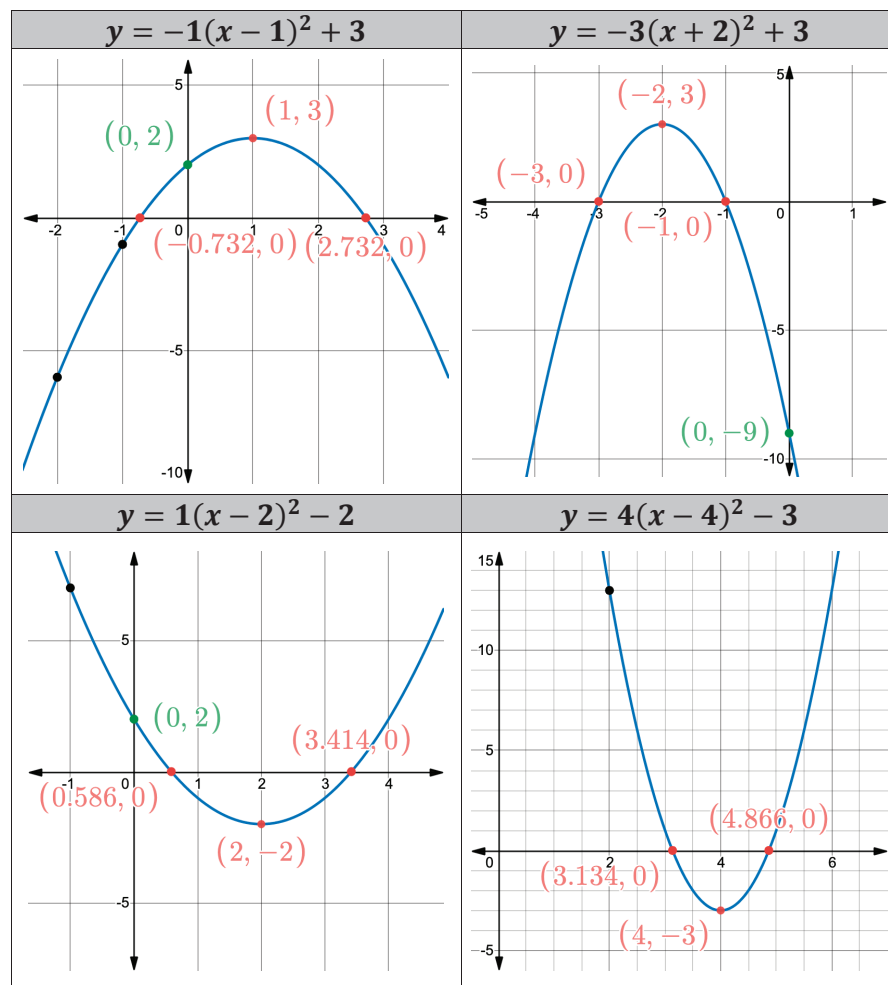
5. The equation of the **red** parabola is $y = -2(x + 3)^2 + 2$. Based on the definition of domain, what values can be used as an input for x for the **red** parabola?
6. The equation of the **purple** parabola is $y = 2(x - 1)^2 + 2$. Based on the definition of domain, what values can be used as an input for x for the **purple** parabola?
7. Make a conjecture about the domain of quadratic functions.



The **range** is the complete set of possible values of the output of a relation or function.

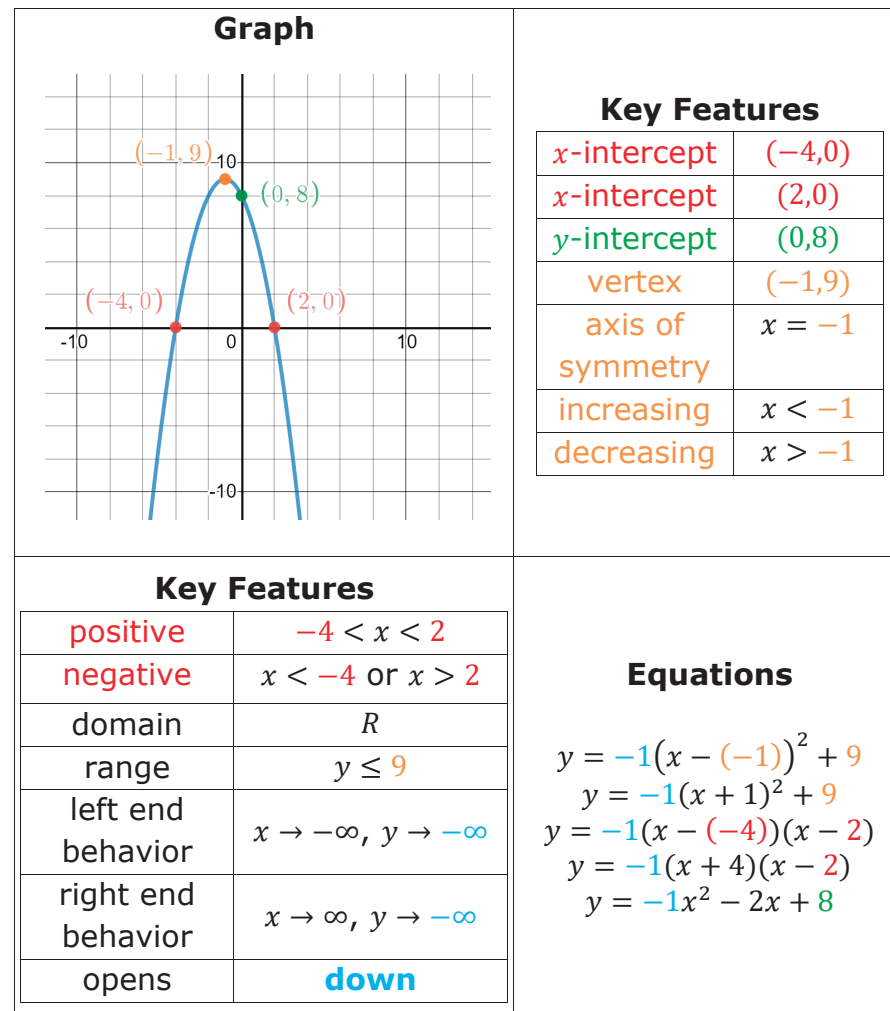
8. The equation of the **red** parabola is $y = -2(x + 3)^2 + 2$. Based on the definition of range, what values are possible for y for the **red** parabola?
9. The equation of the **purple** parabola is $y = 2(x - 1)^2 + 2$. Based on the definition of range, what values are possible for y for the **purple** parabola?
10. Make a conjecture about the range for all quadratic functions.

11. Use the given graphs to discuss with your partner your conjecture about domain and range for all quadratics. If necessary, update your conjectures.



12. Use the graphs to determine if there is a relationship between the equation and whether the parabola opens up or down.

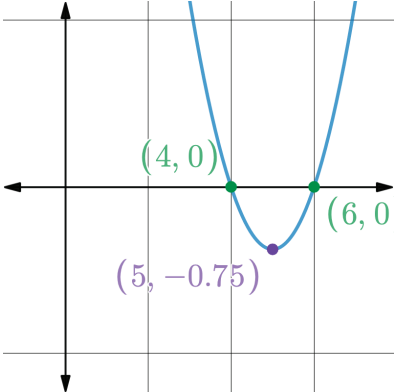
A table and a graph of the quadratic $y = -1(x + 1)^2 + 9$ is shown. The key features of the graph are color coded with the key features in the table. Also, five different forms of the equation of the quadratic are shown with the key features found in each form color coded.



13. How does the fact that the parabola opens down help to determine the end behavior?

14. Use the information in the table to determine the relationship between the x -intercepts and the vertex.

Equations	Graph
$y = -4(x + 7)(x - 5)$ $y = -4(x + 1)^2 + 144$	
x -intercepts	
$(-7, 0)$ and $(5, 0)$	
Vertex	
$(-1, 144)$	

Equations	Graph
$y = \frac{3}{4}(x - 4)(x - 6)$ $y = \frac{3}{4}(x - 5)^2 - \frac{3}{4}$	
x-intercepts	
(4,0) and (6,0)	
Vertex	
$(5, -\frac{3}{4})$	

Equations	Graph
$y = -(x + 11)(x + 3)$ $y = -(x + 7)^2 + 16$	
x -intercepts	
$(-11, 0)$ and $(-3, 0)$	
Vertex	
$(-7, 16)$	

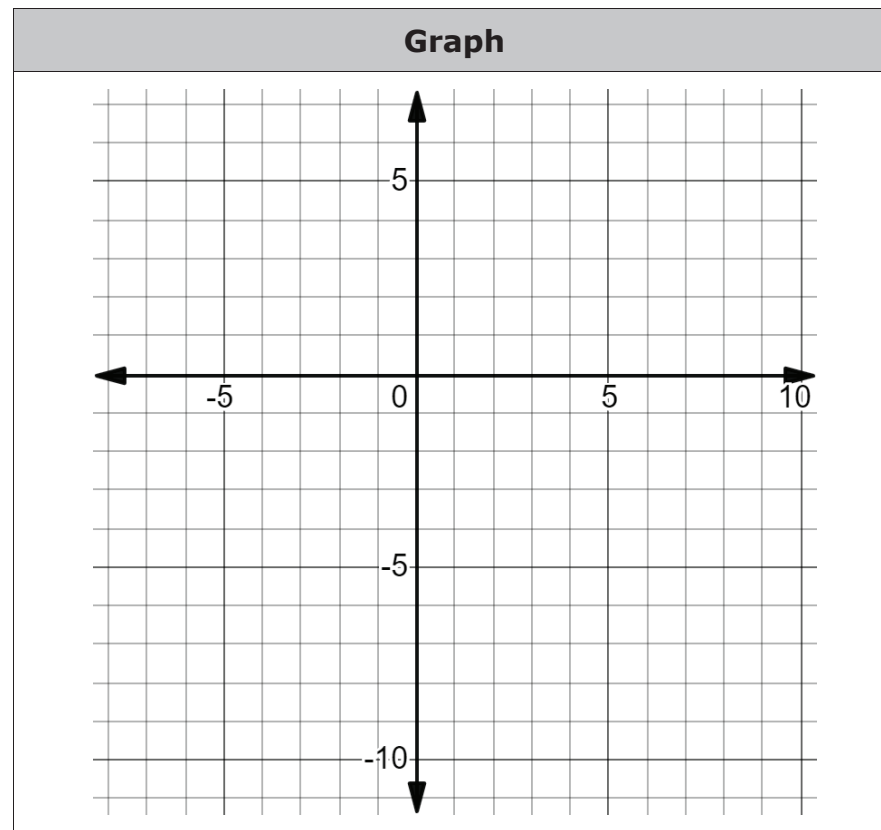
15. The relationship between the x -intercepts and the vertex is



8.2.3 Collaboration: Using Key Features of Quadratics to Graph

- Work with your partner to graph $y = -1(x - 2)^2 + 3$ to complete the following steps.
 - Determine all of the key features **possible** using the equation and table of values.
 - Use the key features to graph the quadratic.

Equation	Table of Values																		
$y = -1(x - 2)^2 + 3$	<table><tr><td>x</td><td>y</td></tr><tr><td>-1</td><td>-6</td></tr><tr><td>0</td><td>-1</td></tr><tr><td>1</td><td>2</td></tr><tr><td>2</td><td>3</td></tr><tr><td>3</td><td>2</td></tr><tr><td>4</td><td>-1</td></tr></table>					x	y	-1	-6	0	-1	1	2	2	3	3	2	4	-1
	x	y																	
	-1	-6																	
	0	-1																	
	1	2																	
	2	3																	
	3	2																	
4	-1																		
Key Features																			
x -intercept		vertex		domain															
x -intercept		increasing		range															
y -intercept		decreasing		opens															
positive		left end behavior	$x \rightarrow -\infty, y \rightarrow \underline{\hspace{2cm}}$																
negative		right end behavior	$x \rightarrow \infty, y \rightarrow \underline{\hspace{2cm}}$																





8.2.4 Your Turn!

- Determine all of the key features **possible** using the equation of the quadratic $y = 4(x + 3)(x - 1)$. Then, use the key features to graph the quadratic. If necessary, complete the table of values.

Key Features					
x-intercept		vertex		domain	
x-intercept		increasing		range	
y-intercept		decreasing		opens	
positive		left end behavior	$x \rightarrow -\infty, y \rightarrow \underline{\hspace{2cm}}$		
negative		right end behavior	$x \rightarrow \infty, y \rightarrow \underline{\hspace{2cm}}$		

Graph	Table of Values																
	<table border="1" style="margin: auto;"> <thead> <tr style="background-color: #d3d3d3;"> <th style="width: 50%;">x</th> <th style="width: 50%;">y</th> </tr> </thead> <tbody> <tr><td> </td><td> </td></tr> <tr><td> </td><td> </td></tr> <tr><td> </td><td> </td></tr> <tr><td> </td><td> </td></tr> <tr><td> </td><td> </td></tr> <tr><td> </td><td> </td></tr> <tr><td> </td><td> </td></tr> </tbody> </table>	x	y														
x	y																



8.2.5 Wrap-Up: Reflection

- Circle the icon that best describes your understanding of each key feature.



I need help.



I got it.



I could help someone.

x-intercepts			
y-intercept			
vertex			
domain			
range			
opens up/down			
increasing/decreasing			
positive/negative			
end behavior			